

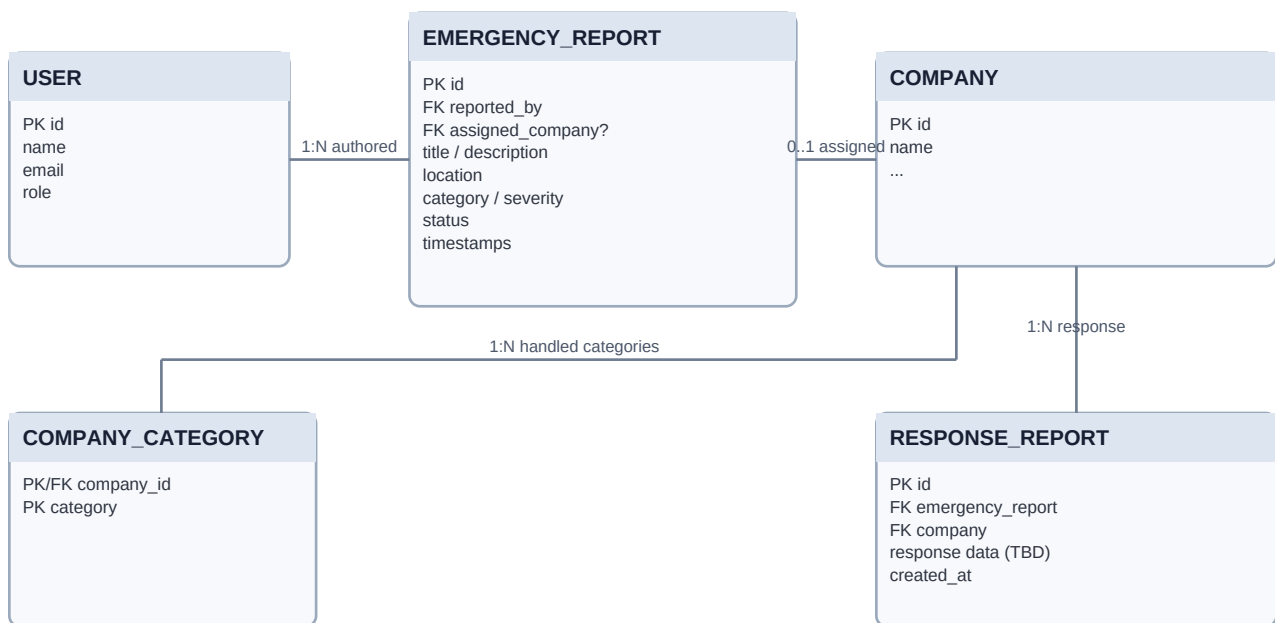
REMIND Lab - Entity-Relational Model

MVP v0.1 | Preliminary logical data model

1. Status

This is a **preliminary logical relational model**, not a database-vendor decision. Its purpose is to formalize relationships and integrity expectations before persistence architecture is selected.

2. ER overview



3. Entity catalog

Entity	Purpose	Key relationships
USER	Represents an authenticated end-user identity and, provisionally, administrative identity.	1:N authored EmergencyReports.
COMPANY	Organization capable of handling configured emergency categories.	1:N categories; 0:N assigned reports; 1:N response records.
EMERGENCY_REPORT	Central occurrence record and lifecycle state.	One author; zero or one assigned Company; response information.
COMPANY_CATEGORY	Associative relation defining categories handled by a Company.	Candidate composite key: company_id + category.
RESPONSE_REPORT	Information recorded by the responsible Company about an attended occurrence.	References EmergencyReport and Company; exact fields/cardinality remain open.

4. Preliminary attributes

Entity	Attribute	Null?	Meaning / constraint
USER	id	No	Primary identifier.
USER	name	No	Profile name.
USER	email	No	Candidate unique identity/contact.
USER	role	No	Provisionally USER or ADMIN.
COMPANY	id	No	Primary identifier.
COMPANY	name	No	Company name.
EMERGENCY_REPORT	id	No	Primary identifier.
EMERGENCY_REPORT	reported_by	No	FK to USER; immutable author.
EMERGENCY_REPORT	assigned_company	Yes	FK to COMPANY; empty before acceptance.
EMERGENCY_REPORT	title	No	Required.
EMERGENCY_REPORT	description	No	Required.
EMERGENCY_REPORT	location	No	Descriptive location.
EMERGENCY_REPORT	category	No	MEDICAL/FIRE/CRIME/ACCIDENT.
EMERGENCY_REPORT	severity	No	Values pending.
EMERGENCY_REPORT	status	No	OPEN/IN_PROGRESS/RESOLVED/CANCELLED.
EMERGENCY_REPORT	created_at	No	System timestamp.
EMERGENCY_REPORT	resolved_at	Yes	Set on resolution.
EMERGENCY_REPORT	cancelled_at	Yes	Set on cancellation.
COMPANY_CATEGORY	company_id	No	FK to COMPANY.
COMPANY_CATEGORY	category	No	Handled category.
RESPONSE_REPORT	id	No	Primary identifier.
RESPONSE_REPORT	emergency_report_id	No	FK to occurrence.
RESPONSE_REPORT	company_id	No	FK to responsible Company.
RESPONSE_REPORT	response data	TBD	Fields intentionally not invented yet.

5. Integrity rules

- Every EmergencyReport references its author.
- A Company may assume only reports matching its configured categories.
- At most one Company is assigned to an EmergencyReport.
- Cancellation is not physical deletion.
- ResponseReport cardinality remains provisional until its requirements are finalized.

6. Open persistence questions

- Unified identity versus dedicated Company account model.
- Whether Admin is a USER role or separate identity.
- Database vendor, identifier strategy, enum mapping and soft-delete strategy.
- Whether lifecycle history requires a dedicated table.